

$$\begin{aligned} \text{a) } f(x) &= \sqrt{x} \\ g(x) &= x+5 \end{aligned}$$

$$(f \circ g)(x) = f(g(x)) = f(x+5) = \sqrt{x+5}$$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{x} + 5$$

$$\begin{aligned} \text{d) } f(x) &= \frac{4}{x^3-6} \\ g(x) &= 2\sqrt[3]{x+1} \\ (f \circ g)(x) &= \frac{4}{(2\sqrt[3]{x+1})^3-6} = \frac{4}{8(x+1)-6} = \\ &= \frac{4}{8x+2} \end{aligned}$$

$$(g \circ f)(x) = 2\sqrt[3]{\frac{4}{x^3-6} + 1}$$

$$\begin{aligned} \text{b) } f(x) &= \cos x \\ g(x) &= x^2 \end{aligned}$$

$$(f \circ g)(x) = f(x^2) = \cos x^2$$

$$(g \circ f)(x) = g(\cos x) = \cos^2 x$$

$$\begin{aligned} \text{e) } f(x) &= \sqrt[3]{x} \\ g(x) &= \frac{1}{x} \\ h(x) &= x^2 \end{aligned}$$

$$f(g(h(x))) = f(g(x^2)) = f\left(\frac{1}{x^2}\right) = \sqrt[3]{\frac{1}{x^2}}$$

$$g(f(h(x))) = g(f(x^2)) = g(\sqrt[3]{x^2}) = \frac{1}{\sqrt[3]{x^2}}$$

$$h(f(g(x))) = h(f\left(\frac{1}{x}\right)) = h\left(\sqrt[3]{\frac{1}{x}}\right) = \left(\sqrt[3]{\frac{1}{x}}\right)^2$$

$$\begin{aligned} \text{c) } f(x) &= \log_2(x-5) \\ g(x) &= 7\cos 3x \end{aligned}$$

$$(f \circ g)(x) = f(7\cos 3x) = \log_2(7\cos 3x - 5)$$

$$(g \circ f)(x) = g(\log_2(x-5)) = 7\cos(3\log_2(x-5))$$

$$f(f(x)) = \log_2(\log_2(x-5)-5)$$

$$\begin{aligned} \text{j) } f(x) &= \log_5 x \\ g(x) &= 5^x \\ h(x) &= x^5 \end{aligned}$$

$$f(g(h(x))) = f(g(x^5)) = f(5^{x^5}) = \log_5 5^{x^5} = x^5$$

$$\begin{aligned} h(g(f(x))) &= h(g(\log_5 x)) = h(5^{\log_5 x}) = \\ &= h(x) = x^5 \end{aligned}$$

$$\boxed{a^{\log_a b} = b}$$